

5 In the diagram, $ABCDE$ is a regular pentagon.

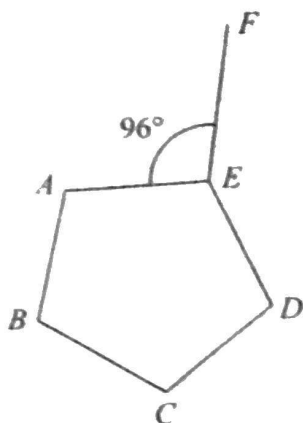


Diagram NOT
accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

$$\begin{aligned} \text{Total Interior } \angle &= 180(n-2) \\ &= 180(5-2) \\ &= 540 \end{aligned}$$

$$\begin{aligned} \therefore \text{Individual interior } \angle &= \frac{540}{5} \\ &= 108 \end{aligned}$$

$$\therefore \angle AED = 108$$

$$96 + 108 + \angle FED = 360 \text{ (}\angle\text{s around a point)}$$

$$204 + \angle FED = 360$$

$$\begin{aligned} \angle FED &= 360 - \cancel{204} \\ &= \underline{156} \end{aligned}$$

156°

(Total for Question 5 is 4 marks)

DO NOT WRITE IN THIS AREA

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